

AskPDF AI Document Intelligence Platform

AskPDF is a production-grade AI Document Intelligence Platform.

Users upload PDF documents and the system processes them using Retrieval Augmented Generation to enable natural language chat.

The platform extracts text from uploaded PDFs, chunks the content into semantically meaningful segments, generates vector embeddings, and stores them for efficient similarity search at query time.

Key Features:

- Secure PDF upload with MIME type validation
- Automatic text extraction using pdf-parse
- Document processing pipeline with status tracking
- MongoDB Atlas for persistent document storage
- RESTful API built with Express and TypeScript

Architecture Overview

The backend follows clean architecture principles:

Routes: Register endpoints and delegate to controllers.

Controllers: Coordinate requests and shape responses.

Services: Contain all business logic and data operations.

Models: Define MongoDB schemas with Mongoose.

Middleware: Handle cross-cutting concerns like file uploads.

Config: Manage database connections and environment setup.

Processing Pipeline:

1. Upload PDF to server storage
2. Create document record in MongoDB (status: uploaded)
3. Extract text from PDF (status: extracting -> extracted)
4. Chunk text into segments (status: chunking -> chunked)
5. Generate embeddings (status: embedding -> ready)
6. Enable AI chat over document content

Technology Stack

Backend: Node.js, Express, TypeScript, Mongoose

Database: MongoDB Atlas

AI: Google Gemini, Google Embeddings

Vector Store: ChromaDB (initial), Pinecone (future)

File Processing: Multer, pdf-parse

Validation: Zod

Authentication: JWT, bcrypt

Frontend: React, TypeScript, Vite, Tailwind CSS

UI Components: shadcn/ui, Framer Motion, Lucide Icons

Data Fetching: TanStack Query, Axios

Forms: React Hook Form

Routing: React Router

This document is a test asset for verifying the PDF upload and text extraction pipeline of the AskPDF platform.